

Valve Inventory

Installation Manual

This covers getting the tool running from nothing - whether you're setting it up for the first time, or you were handed an exported copy by someone else (see File > Export archive and tools in the app, or QUICKSTART.md).

1. Requirements

- **Python 3.8 or later.** Check with `python3 --version`.
- **tkinter**, for the desktop window. Ships with most Python installs; on Debian/Ubuntu it's a separate package: `sudo apt install python3-tk`.
- **openpyxl**, only if you want to export a spreadsheet snapshot: `pip install openpyxl`.
- **reportlab**, only if you want to regenerate these manuals: `pip install reportlab`.

Note: Nothing else is required at runtime. The database engine (SQLite) is built into Python's standard library - there is no server to install or configure.

2. Getting the files

Two starting points:

2a. You have a git clone

```
git clone <repository-url>
cd valve-inventory
```

2b. You were given an exported .zip

Produced by File > Export archive and tools in the GUI. Just unzip it anywhere - it's not tied to git.

```
unzip valve_inventory_export.zip -d valve-inventory
cd valve-inventory
```

3. Building the database

The working database, `valves.db`, is never itself committed or exported - it's a binary SQLite file that would just bloat version control and can't be diffed. What travels with the project is a text snapshot in `data/`, which rebuilds it:

```
python3 snapshot.py --restore
```

This reads `data/valves.sql` and writes a fresh `valves.db` next to it. Safe to re-run any time - it will refuse to overwrite an existing database unless you pass `--force`.

4. Confirming it worked

```
python3 test_smoke.py
```

This runs a fast set of checks - the type-name classifier, a database round-trip, each CLI command against a scratch copy, and (if data/ is present) the restore path itself. It should print all checks passed and exit cleanly. If it doesn't, see Troubleshooting below before going further.

5. Running it

Desktop window

```
python3 valves_gui.py
```

Command line

```
python3 valves.py stats
```

Both read and write the same `valves.db` - there is no separate setup for one versus the other.

6. The datasheet archive (optional)

PDF datasheets are not included in a clone or an export - hundreds of megabytes of third-party files that would swamp the repository. Build your own copy locally:

```
python3 fetch_datasheets.py --index      # maps the site - slow, run once, resumable
python3 fetch_datasheets.py --download   # pulls only the types this database holds
python3 valves.py scan                   # links the downloaded files in
```

The default 2-second delay between requests is deliberate - the source site runs on donations. Both stages are resumable, so Ctrl-C is always safe. If you'd rather have Claude do this work (including finding sources beyond that one site), the GUI's Tools > Generate datasheet download prompt... writes a ready-to-use prompt for an agent with file and web access, such as Claude Code.

7. Troubleshooting

Symptom	Likely cause / fix
"No module named tkinter"	Install the platform tkinter package (e.g. python3-tk on Debian/Ubuntu); it isn't installable via pip.
UnicodeDecodeError / UnicodeEncodeError during restore or CLI output	Windows consoles default to a non-UTF-8 codepage; some collection notes contain Cyrillic and other non-Latin text. This tool's own scripts already force UTF-8 - if you hit this in a modified copy, add <code>encoding="utf-8"</code> to the relevant <code>open()</code> call and <code>sys.stdout.reconfigure(encoding="utf-8")</code> near the top of <code>main()</code> .

Symptom	Likely cause / fix
"valves.db already exists" on restore	Expected safety behaviour - pass --force if you intend to overwrite it, or delete/rename the existing file first if you're not sure what's in it.
GUI window closes immediately, no error visible	Run python3 valves_gui.py from a terminal instead of double-clicking the file, so any traceback stays visible after the window closes.
openpyxl / reportlab import errors	Both are optional - only needed for File > Export spreadsheet and for rebuilding these manuals respectively. pip install the missing one, or simply avoid that feature.